

From Cars to Clinics

Service-Oriented Middleware for Networked Medical Devices

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Cold Open

A patient at home. A clinician in the hospital.

- A wearable sensor on the belt streams 3D acceleration at 100 Hz.
- A clinician's dashboard shows the patient's gait cadence in real time.
- Multiple sclerosis, follow-up, no clinic visit needed today.

One question drives the next 20 minutes

How does the dashboard actually talk to the sensor — safely, interoperably, in real time?

Framing

Prerequisites I assume

From your previous semesters:

Semester	Course	What I will reuse
3	Kommunikationsnetze	OSI, TCP/UDP, HTTP, sockets
3	Multimodale Signalverarbeitung	Sampling, sensor data streams
4	Verteilte Systeme	SOA, multi-tier, security
4	Softwaretechnik	IEC 62304 software lifecycle
4	Standards und Prozesse	Medical workflow vs. informatics

Where we are going

1. Requirements of a networked medical device
2. **Deep dive:** SDC / IEEE 11073
3. Automotive lens — what transfers
4. Where this fits in the wider e-health stack
5. Research directions I would bring to Reutlingen

Requirements

What a networked medical device must offer

1. **Interoperability** — multi-vendor plug-and-play in the OR / at home
2. **Real time** — alarms, closed-loop control, streaming vitals
3. **Safety** — IEC 62304, ISO 14971, IEC 60601-1
4. **Security** — IEC 62443, MDR, threat modelling
5. **Traceability** — who did what, when, from which device
6. **Scalability** — device → room → hospital → network

A single scenario forces all six

- Wearable sends *TidalVolume*, *HeartRate*, *StepCadence*.
- Dashboard subscribes, renders, raises an alert if a threshold is crossed.
- Multiple vendors. Multiple networks. Multiple regulatory jurisdictions.

These are the non-functional requirements. They drive the architecture.

Deep dive — SDC / IEEE 11073

IEEE 11073 — a family, not a single standard

Part	Scope
11073-10207	Domain Information & Service Model (BICEPS)
11073-20701	Service-Oriented Device Connectivity (SDC)
11073-20702	Medical DPWS binding (transport)
11073-10101(nu)	Nomenclature codes for vitals, alarms, events

BICEPS = Basic Integrated Clinical Environment Protocol Specification.

The three roles in SDC

- **Provider** — a medical device that offers services (a ventilator).
- **Consumer** — a system that consumes them (a monitor, a dashboard).
- **SDC network** — discoverable, self-describing, secured.

Everything above the network layer is **domain-specific medical semantics**.

BICEPS in one picture

Every provider hosts an **MDIB** (Medical Device Information Base):

MDIB

Descriptors (what exists: static structure)

MDS (Medical Device System)

VMD (Virtual Medical Device)

Channel

Metric ← e.g., StepCadence

States (current values: dynamic content)

NumericMetricState (single scalar)

StringMetricState

RealTimeSampleArrayState (waveform)

AlertConditionState

AlertSignalState

The four service groups

- **Get Service** — pull the MDIB, one snapshot.
- **Set Service** — change device settings (with safety context!).
- **Description / State Event Service** — push updates to subscribers.
- **Context Service** — who is the patient, which location, which workflow?

Safety Contexts — the hard part

A *Set* operation is not just an RPC. The consumer must include a **Safety Context** proving:

- the right **patient** is targeted (PatientContext),
- the right **location** (LocationContext),
- the right **workflow** (WorkflowContext),
- and that the operation is **permitted right now** (Participant Keys).

This is where SDC earns the word "medical".

- **WS-Discovery** over UDP multicast — how devices find each other
- **SOAP/HTTP(S)** over TCP — for messages
- **WS-Eventing** — for subscriptions
- **TLS** — transport security
- **WS-Security + XML Signature** — message integrity above TLS

Layering, cleanly separated.

Sequence: discovery → subscribe → stream

Consumer	Provider (wearable)
--- Probe (WS-Discovery) ----->	
<-- ProbeMatch -----	
--- GetMdib ----->	
<-- MDIB (descriptors + states)--	
--- Subscribe (State Events) --->	
<-- SubscribeResponse -----	
<== EpisodicMetricReport =====	(StepCadence = 108/min)
<== EpisodicMetricReport =====	(StepCadence = 106/min)
<== EpisodicAlertReport =====	(Cadence below threshold)

Look at the corner of your screen

The dashboard from the opening slide has been running through this whole section.

Provider streams synthetic 3D-accelerometer data. Consumer computes rolling cadence. No cloud. No configuration. Localhost only.

We will come back to this at the end.

Automotive lens

Same problem, different domain

In eight years at Bosch I built middleware for a car with 100+ ECUs. The requirements list I showed you earlier?

Word for word the same, minus one column.

Concern	Automotive	Medical
Service-oriented comms	SOME/IP (AUTOSAR AP)	SDC (IEEE 11073-20701)
Real-time streaming	DDS, iceoryx (zero-copy)	DDS, SDC waveform states
Safety lifecycle	ISO 26262 (ASIL A-D)	IEC 62304 (Class A/B/C) + ISO 14959
Cybersecurity	ISO/SAE 21434	IEC 62443 + MDR
Process maturity	Automotive SPICE	ISO 13485 + IEC 62304
Risk acceptance	Residual risk only	Residual risk vs. clinical benefit

What transfers, what does not

Transfers well:

- Service-oriented architectural patterns
- QoS-based real-time streaming
- Risk-based V-model and traceability discipline
- Test infrastructure and simulation (my digital-twin work at Bosch)

Does *not* transfer:

- **Domain semantics** — BICEPS is medicine, VLANs are cars
- **Clinical benefit** in the risk equation — unique to medicine
- **Patient / workflow context** — no analogue in the vehicle

Architectural patterns are transferable. Domain semantics are not.

That is where the real engineering happens. And that is what I want to teach and research at Reutlingen.

Where SDC fits

One wearable, one dashboard — but there is more

```
+-----+
| Physician office |
| / EHR / national |
| telematics platform |
+-----+-----+
          | FHIR / TI-Messenger / KIM
          | (system-to-system)
+-----+-----+
| Hospital gateway |
+-----+-----+
          | SDC / IEEE 11073
          | (device-to-device)
+-----+-----+
| Bedside monitor, |
```

Two layers of the same stack

- Above the device layer: the **Telematikinfrastuktur** — topic of my German talk earlier today.
- Below the device layer: *this* talk.

A "networked medical system" is not one protocol. It is a consistent architectural stance across many.

Research at Reutlingen

What I would bring to CaMed

Three concrete directions — each with an existing anchor at Reutlingen:

1. **SDC as the substrate for the connected OR** — image-guided procedures, real-time device orchestration, 5G-connected surgical scenarios.
2. **Digital-twin-based V&V of networked device configurations** — carrying over the Bosch/ETH work on automotive digital twins.
3. **Uncertainty quantification for sensor-fusion pipelines** — from wearables to closed-loop clinical decisions (my PhD, applied to medicine).

- Project-based, end-to-end: requirements → architecture → code → V&V → regulatory dossier.
- Everything as code, everything reproducible, everything reviewable.
- This lecture is the smallest example: 20 minutes of talk, one git repository.

Closing the circle

The dashboard you saw at minute one

- Was running from this repository.
- Provider + consumer + dashboard in ~200 lines of Python.
- Synthetic gait data — deterministic, reproducible, CC0.

These slides were written in Org, rendered as reveal.js, versioned in Git. The Beamer PDF was produced by the same source, via Pandoc, in CI.

These slides. This demo. Reproducible. Public domain.

Any of you can run make demo tonight.

Thank you — questions?

Backup slides

BICEPS metric types in detail

- `NumericMetricState` — single scalar with unit + timestamp
- `StringMetricState` — textual value
- `EnumStringMetricState` — enumerated value
- `RealTimeSampleArrayMetricState` — waveform, sampled data
- `DistributionSampleArrayMetricState` — statistical distributions

- Software development plan
- Software requirements specification
- Software architecture
- Detailed design
- Unit implementation + verification
- Integration + integration testing
- System testing
- Release documentation
- Problem resolution + change control across the lifecycle

SDC vs. FHIR — one line

- **SDC**: real-time, device-to-device, inside a clinical space, milliseconds.
- **FHIR**: system-to-system, cross-institution, over the internet, seconds.

They are complementary, not competing.